

SMART INDIA HACKATHON 2026 — 5-MINUTE PRESENTATION SCRIPT

PROJECT A.U.R.A. // TEAM DELTA — Panimalar Engineering College (Autonomous)

Panel Lead: Dr. S. Balaji (Professor, Dept of CSE)
Schedule: 7.9.2026, 6:00 PM – 8:00 PM
Platform: Google Meet (gad-jeft-gtj)

Team Name: TEAM DELTA
Problem Statement ID: SIH1566
Speakers: Kingston J & Nithyashree V

■■ SLIDE 1: OFFICIAL INTRODUCTION & TEAM DETAILS

■■ [0:00 – 0:45] SPEAKER: KINGSTON J

(Action: Camera ON. Slide 1 visible with SIH Logo and college header. Kingston speaks clearly and formally)

"Good evening respected Panel Head Dr. Balaji Sir and honorable judges.

We are **Team DELTA** representing **Panimalar Engineering College (Autonomous)**.

We are presenting for **Problem Statement ID: SIH1566**, under the theme **Disaster Management and Robotics**, categorized as a **Hardware and Software Integrated System**.

Our Problem Statement Title is: *'Rapid Non-Invasive Subterranean Victim Localization & Void Life Detection in Post-Disaster Structural Collapse Scenarios.'*

My name is **Kingston J**, and presenting along with me is my teammate **Nithyashree V**. Together with our team members Jaffer Rilwaan, Hannah Blessy, Kathiravan, and Giridhar, we have engineered **Project A.U.R.A. — the Autonomous Universal Rescue & Analysis Node**."

■■ SLIDE 2: THE GROUND REALITY & THE 72-HOUR GOLDEN WINDOW

■■ [0:45 – 1:45] SPEAKER: KINGSTON J

(Action: Switch to Slide 2. Kingston changes tone to serious, compelling disaster storytelling)

"Let me take you to ground zero of an earthquake or a devastating landslide like Wayanad.

When a multi-story building collapses into pancake rubble layers, human beings get trapped in air pockets beneath tons of reinforced concrete. Medical science tells us that trapped victims have only **72 hours — the Golden Window** — before dehydration, dust suffocation, and toxic gases end their lives.

When our NDRF and SDRF teams arrive, they face an impenetrable wall. Flying drones with thermal cameras are useless because thermal infrared cannot see through concrete slabs. Walkie-talkies and phone signals are completely dead under dense steel rebar.

Existing military equipment used by defense forces weighs **15 to 20 kilograms**, costs an extreme **15 Lakh Rupees**, and can only penetrate **3 to 4 meters** in simple brick walls.

Because rescue teams have no vision, excavators dig blindly. That blind digging destabilizes the rubble arches, causing secondary cave-ins that crush the very people teams are trying to save.

We asked a simple question: *Why should lives be lost just because our rescue sensors cannot penetrate concrete?* That is why we built Project A.U.R.A."

■■ SLIDE 3: HOW PROJECT A.U.R.A. WORKS & OUR TESTED SENSOR RANGES

■■ [1:45 – 3:00] SPEAKER: NITHYASHREE V

(Action: Switch to Slide 3. Nithyashree holds up the physical prototype probe to the webcam)

"Thank you, Kingston. Respected judges, think of Project A.U.R.A. as a doctor's stethoscope, an acoustic ear, a radar eye, and a trained search dog — combined into one lightweight, handheld node.

Instead of guessing, A.U.R.A. combines four distinct physical sensors powered by an ESP32 dual-core computer running FreeRTOS:

- 1. Our 3.18 GHz Microwave Doppler Radar on Pin 33:** While camera light bounces off concrete, radio microwaves travel right through drywall, brick, and rubble up to **2.5 meters deep** — and up to **6.0 meters in open void air**. It detects the unconscious rise-and-fall chest breathing and body twitching of a victim.
- 2. Our Piezoelectric Seismic Geophone on Pin 35:** It presses directly against concrete slabs. When a trapped victim knocks with a stone or pipe, it captures structural shockwaves traveling through steel pipes and rebar up to **8.0 meters away**.
- 3. Our Low-Noise Acoustic Microphone on Pin 32:** It listens along rubble cracks and air fissures between **2.0 to 5.0 meters**, sampling sound 500 times a second to detect faint whispers, gasps, and crying.
- 4. Our MQ-135 Bio-Scent Gas Sensor on Pin 34:** It acts like a digital search dog, detecting exhaled carbon dioxide, ammonia, and human sweat VOCs within a **1.0 to 2.5 meter diffusion radius**, while warning teams of hazardous toxic gases.

Finally, an MPU-6050 sensor monitors debris shifting to protect responders from aftershocks, and an onboard 8-slide OLED screen lets firefighters read victim depth directly on the hardware."

■■ SLIDE 4: HOW DEPTH IS CALCULATED & THE 3 COLOR TRIAGE ZONES

■■ [3:00 – 4:00] SPEAKER: KINGSTON J

(Action: Switch to Slide 4. Kingston explains the calculation simply and points to the color zones)

"Now, how does A.U.R.A. calculate where to dig?

It uses a well-proven law of physics called energy attenuation: the further sound and shockwaves travel through concrete, the weaker they become. By combining the acoustic energy and seismic impulse peaks, A.U.R.A. computes victim distance using the formula:

Distance = 12 divided by square root of max energy, multiplied by 0.8 for true depth.

To make this instant and foolproof for firefighters on the ground, A.U.R.A. translates this into **Three Simple Color Zones** on the screen:

- **GREEN ZONE (Under 0.5 meters):** The survivor is located directly under surface rubble. Responders can dig immediately with hand shovels and crowbars.
- **RED ZONE (0.5 to 2.5 meters):** The victim is buried under structural slabs. Rescuers must bring hydraulic spreaders, jacks, and concrete cutters.
- **WHITE ZONE (Over 2.5 meters, up to 6.0 meters):** The victim is deep down in a basement cavity. Teams must shore up the walls and drill a narrow oxygen pipe first.

Every single measurement is genuine physical sensor data. Our ESP32 streams this over WiFi directly into **Google Gemini Flash AI**, which gives the commander a clean directive: *'Survivor located 2.4 meters deep at 320 degrees North-Northwest.'*"

■■ SLIDE 5 & 6: COST ECONOMICS, HONEST LIMITS & CONCLUSION

■ ■ [4:00 – 4:45] SPEAKER: NITHYASHREE V

(Action: Switch to Slide 5/6. Show live dashboard running at <http://localhost:5173>)

"As you can see on our live screen right now, our physical working prototype is online and streaming live data to our React 19 Tactical C2 Dashboard.

Now let us look at the Economics and Feasibility:

An imported foreign search system like Delsar costs **12 to 15 Lakh Rupees**. Because of that price, only elite central units have them. District and local fire stations have zero.

Project A.U.R.A. is built at **under ■3,500 Rupees — less than 42 US Dollars**. That means for the price of ONE imported machine, India can equip **400 fire stations and rescue squads** across every district in our country.

We also believe in honest engineering: **What are our limits?**

A.U.R.A. easily penetrates concrete slabs 20 to 30 cm thick, brick walls, and dry rubble. However, it cannot penetrate solid sheet-metal or flooded waterlogged mud. But because we combine radar, tapping, sound, and scent, when one sensor hits a barrier, the other three take over.

Our Phase 2 is already tested: adding LoRa mesh swarm networking so multiple A.U.R.A. nodes communicate across an entire disaster field simultaneously."

■ ■ [4:45 – 5:00] BOTH TOGETHER — GRAND FINALE CLOSING

KINGSTON: "Project A.U.R.A. is engineered to hear the unheard and bring hope into the darkest rubble."

NITHYASHREE: "Representing Panimalar Engineering College, we are **Team DELTA**, and our working prototype is live and ready for your evaluation. Thank you, and we welcome your questions!"

■ ■ 3-MINUTE JUDGES' Q&A; SURVIVAL GUIDE

Q1: How do you prove your sensor data is real and not simulated?

Kingston: 'Sir, our prototype is streaming right now. If I tap the desk near our geophone, the impulse spikes, the tap counter increments, and decibels rise. In a quiet room, the acoustic spectrum reads SILENCE and victim count is zero. Every value comes directly from ESP32 ADC1 registers (Pins 32, 34, 35) with zero synthetic floors.'

Q2: Why does microwave radar penetrate concrete when optical drones fail?

Nithyashree: 'Optical cameras use visible light with nanometer wavelengths absorbed by concrete surfaces. Our RCWL-0516 uses 3.18 GHz microwave radio waves. These long radio waves travel right through non-metallic dielectric materials like concrete, brick, and wood, reflecting off the dielectric contrast of moving human chest muscle.'

Q3: What if there is no Wi-Fi or internet in a collapsed disaster area?

Kingston: 'A.U.R.A. is 100% offline-first. The ESP32 hosts its own local Wi-Fi hotspot and web server. Responders connect directly without internet. Furthermore, the onboard 8-slide OLED screen lets a firefighter read depth, tap count, and gas levels directly on the probe even if all tablets fail.'

Q4: How did you keep the cost under Rs. 3,500?

Nithyashree: 'Foreign systems use proprietary military FPGA chips. We replaced costly hardware with software — running our 500Hz DSP on Core 1 of an ESP32-WROOM-32. Using commercial off-the-shelf sensors keeps our total component bill to exactly Rs. 3,420 INR.'